

A proof of Barry's Hankel–Somos-6 conjecture

Polynomial specialization, a homogeneous family,
and reproducible exact verification

AI-assisted research note

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Abstract

Conjecture 7 of Paul Barry's 2022 paper proposes a three-parameter family of quadratic algebraic generating functions whose Hankel determinants satisfy a Somos-6 recurrence with its middle coefficient zero. We prove the claimed bilinear identity and its homogeneous four-parameter form. The decisive observation is a constant square completion: the apparent cubic remainder in the associated sextic discriminant becomes linear. This identifies the recurrence coefficients with the invariants in van der Poorten's 2005 continued-fraction theorem. We give an algebraic derivation of the needed local identity and the Hankel correspondence.

A genuine technical issue is that the original family has an exceptional initial complete quotient, and some parameter values produce zero Hankel determinants. We handle both by embedding the family in a polynomial seven-parameter deformation. An explicit Catalan specialization proves that all its generic Hankel determinants are nonzero. The resulting identity is then specialized without division. Consequently the final statement holds over every commutative coefficient ring, including at singular parameters. The accompanying programs independently compute determinants and verify 10,248 recurrence residuals in 1,128 test cases, with additional example and symbolic checks.

Main coefficient simplification. For Barry's parameters,

$$\alpha = t^2(r + 2)^2, \quad \gamma = t^3((s + t)(r + s + t + 2)^2 + t(r + 2)^3).$$

The second expression expands to the polynomial in Conjecture 7.

Status and attribution. This is an unrefereed research manuscript, not a machine-checked formalization. The genus-two continued-fraction identity is due to van der Poorten; it is not claimed as a new result here. The contribution is the explicit reduction, the complete specialization argument, and a proof of the named conjecture. A targeted literature search did not locate an earlier direct resolution of Conjecture 7, but does not establish a priority claim.

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Reading guide

The problem and answer are in Sections 1–2. Section 4 isolates the algebraic observation that makes the problem tractable. Sections 5–7 contain the all-index proof, including the handling of zero determinants. Sections 8–10 provide examples, consequences, and a reproducibility audit.

1 The selected problem and its status

Let $G(x) = \sum_{n \geq 0} g_n x^n$ be the formal power series with $G(0) = 1$ determined by

$$G(x) = \frac{1}{1 - x \frac{1 + rx}{1 - x} - x^2 \frac{1 + sx}{1 - x} - tx^3 G(x)}. \quad (1.1)$$

Barry [1, Conjecture 7, p. 7] conjectured that the Hankel transform of its coefficients is an $(\alpha, 0, \gamma)$ Somos-6 sequence. The absence of the middle term means that the identity has only three terms, despite its index width being six. In the general Somos-6 recurrence there is an additional term $\beta H_{N-2} H_{N-4}$; the conjecture asserts $\beta = 0$.

We use matrix size as the determinant index:

$$H_0 = 1, \quad H_N = \det(g_{i+j})_{0 \leq i, j < N} \quad (N \geq 1). \quad (1.2)$$

Barry's indexing is $h_n = H_{n+1}$. Keeping this shift explicit avoids a common source of mistakes in comparisons of Hankel recurrences.

Theorem 1.1 (Barry's Conjecture 7, in polynomial form). *For the series (1.1), the determinants (1.2) satisfy*

$$H_N H_{N-6} = \alpha H_{N-1} H_{N-5} + \gamma H_{N-3}^2 \quad (N \geq 6), \quad (1.3)$$

where

$$\alpha = t^2(r+2)^2, \quad \gamma = t^3((s+t)(r+s+t+2)^2 + t(r+2)^3). \quad (1.4)$$

This is an identity in $\mathbb{Z}[r, s, t]$ and hence remains true under every specialization into a commutative ring. Over a field, whenever $H_{N-6} \neq 0$, (1.3) is the usual quotient-form Somos recurrence.

With $h_n = H_{n+1}$, the conjectured identity follows for $n \geq 6$. The theorem also supplies the preceding boundary relation corresponding to $N = 6$, with the empty determinant $H_0 = 1$.

1.1 What was checked in the literature

The source problem is a specific displayed conjecture, not an artificially constructed question. Wang and Zhang [4] proved the four Somos-4 conjectures numbered 2–5 in Barry's paper; their Corollaries 4–7 identify those numbers explicitly. That result should not be mistaken for a proof of the Somos-6 conjecture considered here.

The central earlier theorem is van der Poorten's width-six relation [2, Theorems 3.1–3.2]. Hone's treatment [3, Section 4 and Theorem 5.4] places continued fractions and Hankel determinants in a broader hyperelliptic framework. Those results made the problem a plausible target: the missing task was to identify the particular curve and determinant normalization, rather than to create an entire theory of Somos sequences.

The search was checked against recent related work as well. The stated applications in Chang–Liu [5] concern bilateral Somos-4 and Somos-5, and Scheuerle's September 2026 preprint [6] concerns Somos-4. No direct resolution of Barry's Conjecture 7 was located. This is a bounded literature-search statement, not a certification that no unpublished or differently phrased proof exists. The proof below stands independently of that bibliographic uncertainty.

2 A homogeneous formulation of the answer

It is convenient to replace $r+2$ and $s+t$ by independent parameters and to allow a scale parameter u . Define

$$D(x) = 1 - 2ux + (u^2 - R)x^2 + (t - B)x^3 \quad (2.1)$$

and let $F(x) = \sum_{n \geq 0} f_n x^n$, $f_0 = 1$, solve

$$D(x)F(x) = 1 - ux + tx^3(1 - ux)F(x)^2. \quad (2.2)$$

The coefficient of the new unknown f_n is always 1, so this equation uniquely defines F over $\mathbb{Z}[u, R, B, t]$, or over any ring after specialization. Write $H_N(F) = \det(f_{i+j})_{0 \leq i, j < N}$ and $H_0(F) = 1$.

Theorem 2.1 (Homogeneous Hankel–Somos-6 identity). *For all $N \geq 6$,*

$$\begin{aligned} H_N(F)H_{N-6}(F) &= t^2 R^2 H_{N-1}(F)H_{N-5}(F) \\ &\quad + t^3 (B(B + uR)^2 + tR^3) H_{N-3}(F)^2. \end{aligned} \quad (2.3)$$

The equality holds in $\mathbb{Z}[u, R, B, t]$ without any nonvanishing hypotheses.

The substitution

$$u = 1, \quad R = r + 2, \quad B = s + t \quad (2.4)$$

turns (2.2) into (1.1), after multiplication by $1 - x$. Thus theorem 2.1 proves theorem 1.1.

Some useful initial values are

$$\begin{aligned} f_0 &= 1, & f_1 &= u, & f_2 &= u^2 + R, \\ f_3 &= u^3 + 3uR + B, & f_4 &= u^4 + 6u^2R + R^2 + 3uB. \end{aligned} \quad (2.5)$$

Direct determinants give

$$\begin{aligned} H_0 &= H_1 = 1, & H_2 &= R, & H_3 &= -B(B + uR), \\ H_4 &= -t(B(B + uR)^2 + tR^3). \end{aligned} \quad (2.6)$$

These formulas already show why it would be incorrect to assume that all specialized determinants are nonzero or positive.

The fourth parameter is a homogeneous extension, not a claim of an independent new moduli direction: when u is invertible, it can be removed by scaling x . Its inclusion makes the weights and the exceptional case $u = 0$ transparent.

3 Formal coefficients and a combinatorial interpretation

3.1 A triangular coefficient recurrence

Set $f_j = 0$ for $j < 0$ and

$$C_m(F) = \sum_{j=0}^m f_j f_{m-j} \quad (m \geq 0), \quad C_m(F) = 0 \quad (m < 0).$$

Equating coefficients in (2.2) yields

$$\begin{aligned} f_n &= \mathbf{1}_{n=0} - u\mathbf{1}_{n=1} + 2uf_{n-1} + (R - u^2)f_{n-2} + (B - t)f_{n-3} \\ &\quad + tC_{n-3}(F) - uC_{n-4}(F). \end{aligned} \quad (3.1)$$

For $n \geq 1$, only previously determined coefficients occur. In particular, all moments and all Hankel determinants are integer polynomials in the parameters. This elementary observation will ultimately permit division-free specialization of the more complicated continued-fraction argument.

3.2 A finite Catalan formula

Let $C(z) = \sum_{j \geq 0} C_j z^j$ be the Catalan series, characterized formally by $C(z) = 1 + zC(z)^2$, where $C_j = \frac{1}{j+1} \binom{2j}{j}$. Substitution into (2.2) gives

$$F(x) = \frac{1 - ux}{D(x)} C\left(\frac{tx^3(1 - ux)^2}{D(x)^2}\right). \quad (3.2)$$

No square-root branch or analytic convergence convention is needed. The Catalan form in Barry's original problem is also displayed in [1, Conjecture 7].

For example, (3.2) gives the finite coefficient formula

$$f_n = \sum_{j=0}^{\lfloor n/3 \rfloor} C_j t^j [x^{n-3j}] \frac{(1 - ux)^{2j+1}}{D(x)^{2j+1}}. \quad (3.3)$$

One can remove coefficient extraction entirely. In the following expression, $m = 2j + 1$ and $(m)_L = m(m+1) \cdots (m+L-1)$, with $(m)_0 = 1$:

$$f_n = \sum_{j=0}^{\lfloor n/3 \rfloor} C_j t^j \sum_{\substack{0 \leq q \leq m, \ell_1, \ell_2, \ell_3 \geq 0 \\ q + \ell_1 + 2\ell_2 + 3\ell_3 = n - 3j}} \binom{m}{q} (-u)^q \frac{(m)_{\ell_1 + \ell_2 + \ell_3}}{\ell_1! \ell_2! \ell_3!} \cdot (2u)^{\ell_1} (R - u^2)^{\ell_2} (B - t)^{\ell_3}. \quad (3.4)$$

This follows by expanding the numerator binomially and the denominator by the negative multinomial theorem. The displayed factorial ratio is an integer multinomial coefficient. Thus this is also a direct finite formula for the integer sequences in the family.

3.3 Colored unary–binary trees for the original parameters

The original equation can be written in a particularly simple form:

$$G = 1 + W(x)G + tx^3G^2, \quad W(x) = x + (r+2)x^2 + (r+s+2)\frac{x^3}{1-x}. \quad (3.5)$$

For nonnegative integers r, s, t , interpret a tree recursively as an empty object, a unary root with one subtree, or a binary root with two ordered subtrees. A binary root has size 3 and one of t colors. A unary root of size 1 has one color; one of size 2 has $r+2$ colors; one of any size $\ell \geq 3$ has $r+s+2$ colors. The total size is the sum of root sizes. All nonempty roots have positive size, so every coefficient counts a finite set of objects. The recursive construction has exactly (3.5) as its generating equation.

This gives a concrete combinatorial source of the moments. It does not make the Hankel determinants positive: determinants involve signs, and the negative values in (2.6) are entirely compatible with positive moment counts. The verification code also expands (3.5) independently and compares it with (3.1).

4 The decisive square completion

Put $X = x^{-1}$ and introduce the cubic

$$S(X) = X^3 - 2uX^2 + (u^2 - R)X - B + t. \quad (4.1)$$

The discriminant of the quadratic equation for F corresponds to the sextic

$$Y^2 = S(X)^2 - 4tX(X-u)^2. \quad (4.2)$$

At first sight, the remainder next to the square is cubic. The appropriate square, however, is not S^2 . Set

$$A(X) = S(X) - 2t = X^3 - 2uX^2 + (u^2 - R)X - B - t. \quad (4.3)$$

A direct expansion gives

$$\boxed{S(X)^2 - 4tX(X-u)^2 = A(X)^2 - 4t(RX + B)}. \quad (4.4)$$

Indeed, $S(X) - X(X-u)^2 = -RX - B + t$, so subtracting the two squares leaves $-4t(RX + B)$.

Generically, put $v = tR$ and $w = -B/R$. Then the same curve is

$$Y^2 = A(X)^2 - 4v(X-w), \quad Z = \frac{A+Y}{2}, \quad Z^2 - AZ + v(X-w) = 0. \quad (4.5)$$

The invariant coefficients in the width-six continued-fraction identity are v^2 and $-v^3A(w)$. In our case,

$$v^2 = t^2R^2, \quad -v^3A(w) = t^3(B(B + uR)^2 + tR^3). \quad (4.6)$$

This explains both the compact coefficient formula and the disappearance of the middle Somos-6 term. The curve is generically of genus two, but no smoothness assumption is required in the final polynomial identity.

There is also an explicit identification of the moment function. Choose $Y = A + O(X^{-2})$ at infinity (the square-root branch with leading term X^3), and put $M(X) = X^{-1}F(X^{-1})$. The quadratic formula, or direct substitution, gives

$$M(X) = \frac{S - Y}{2t(X - u)} = \frac{X(X - u)}{Z + t}. \quad (4.7)$$

For the second equality, use the norm identity

$$(Z + t)(\bar{Z} + t) = t(A + t) + t(RX + B) = tX(X - u)^2. \quad (4.8)$$

Here the bar denotes the other root of the quadratic equation; it is not complex conjugation.

The natural initial complete quotient is therefore

$$\frac{1}{M} = \frac{Z + t}{X(X - u)}.$$

Its numerator correction is constant. In the customary notation $P_0 = d_0(X + e_0)$, this is a limiting situation with $d_0 = 0$ rather than an ordinary generic initial value. One must not simply invoke a theorem that assumes all the d_j are nonzero and then silently ignore this issue. Moreover, $w = -B/R$ is undefined when $R = 0$. The next two sections resolve both problems by proving a polynomial identity in a larger family first.

5 A polynomial Hankel form of the genus-two identity

Work initially over the polynomial ring

$$\mathcal{S} = \mathbb{Z}[a, b, c, p, k, \lambda, \mu].$$

Define

$$A = X^3 + aX^2 + bX + c, \quad P = pX + k, \quad Q = X^2 + \lambda X + \mu. \quad (5.1)$$

Divide the polynomial $P(A + P)$ by the monic polynomial Q and write the result as

$$P(A + P) + vX + m = QL. \quad (5.2)$$

Thus $-(vX + m)$ is the remainder, and $L = pX^2 + \ell_1X + \ell_0$ is the quotient. Everything in (5.2) is polynomial over $\mathcal{S}$. For implementation, convenient formulas are

$$\begin{aligned} \ell_1 &= ap + k - \lambda p, \\ \ell_0 &= ak - a\lambda p + bp - k\lambda + \lambda^2 p - \mu p + p^2, \\ v &= \ell_1\mu + \ell_0\lambda - cp - bk - 2pk, \\ m &= \ell_0\mu - ck - k^2. \end{aligned} \quad (5.3)$$

Let $\Phi(x) = \sum_{n \geq 0} \phi_n x^n$, $\phi_0 = 1$, be defined by

$$\begin{aligned} (1 + ax + (b + 2p)x^2 + (c + 2k)x^3)\Phi &= 1 + \lambda x + \mu x^2 \\ &\quad + x^2(p + \ell_1x + \ell_0x^2)\Phi^2. \end{aligned} \quad (5.4)$$

Equivalently, $\mathcal{M}(X) = X^{-1}\Phi(X^{-1})$ is the unique Laurent series $X^{-1} + O(X^{-2})$ satisfying

$$Q - (A + 2P)\mathcal{M} + L\mathcal{M}^2 = 0. \quad (5.5)$$

Theorem 5.1 (Polynomial Hankel–Somos identity). *Let $K_N = \det(\phi_{i+j})_{0 \leq i,j < N}$ and $K_0 = 1$. For every $N \geq 6$,*

$$K_N K_{N-6} = v^2 K_{N-1} K_{N-5} + \Gamma K_{N-3}^2, \quad (5.6)$$

where

$$\Gamma = m^3 - avm^2 + bv^2m - cv^3. \quad (5.7)$$

The identity is valid in $\mathcal{S}$, and hence after any commutative-ring specialization. No nonvanishing assumption belongs to the conclusion.

The generic continued-fraction identity behind this theorem is van der Poorten's [2]. The polynomial family and the argument below make its use at exceptional initial data explicit. In particular, the proof does not assume, without verification, that Barry's smaller family meets a suitable generic open set.

6 Proof of the polynomial theorem

6.1 A single specialization proves generic nonvanishing

Every ϕ_n , and therefore every K_N , belongs to $\mathcal{S}$ by the triangular coefficient recursion in (5.4). To prove that none of these polynomials is identically zero, specialize

$$(a, b, c, p, k, \lambda, \mu) = (0, -3, 0, 1, 0, 0, -1). \quad (6.1)$$

Here $L = X^2 - 1$, $v = 0$, and $m = 1$. Equation (5.4) becomes

$$(1 - x^2)\Phi = (1 - x^2) + x^2(1 - x^2)\Phi^2,$$

so $\Phi = C(x^2)$.

For completeness, all Hankel determinants of the interleaved Catalan sequence are 1. Let J be the adjacency matrix of the path on vertices $0, 1, 2, \dots$. Products of this infinite matrix are harmless here: all vectors used have finite support. A walk of length $2j$ from 0 back to 0 is a Dyck path, so

$$(J^{2j})_{00} = C_j, \quad (J^{2j+1})_{00} = 0.$$

Define monic polynomials $\pi_0 = 1$, $\pi_1 = X$, and $\pi_{n+1} = X\pi_n - \pi_{n-1}$. Induction gives $\pi_n(J)e_0 = e_n$. For the moment functional $\mathcal{L}(q) = e_0^T q(J)e_0$, symmetry of J now gives

$$\mathcal{L}(\pi_i \pi_j) = \delta_{ij}.$$

The change from monomials to the monic π_i is unit lower triangular. Consequently the Gram determinant in the monomial basis is also 1. This proves $K_N = 1$ under (6.1), for every N .

It follows that each K_N is nonzero in the fraction field $\mathcal{K} = \mathbb{Q}(a, b, c, p, k, \lambda, \mu)$. Notice that the specialization has $v = 0$; that is no obstacle. Its sole purpose is to prove that the Hankel polynomials are nonzero. The polynomial v itself is not identically zero, as is evident from (5.3), so v is invertible in $\mathcal{K}$. No positivity assertion is being made for generic moments.

6.2 The Jacobi coefficients are Hankel ratios

We recall the precise formal correspondence, with the indexing required here. Suppose $\mathcal{M} = X^{-1} \sum_{j \geq 0} \phi_j X^{-j}$ and all K_N are nonzero. Define a functional by $\mathcal{L}(X^j) = \phi_j$. The linear equations for the monic degree- n polynomial orthogonal to $1, X, \dots, X^{n-1}$ have coefficient matrix with determinant K_n , so they have a unique solution π_n . If $v_n = \mathcal{L}(\pi_n^2)$, a monic change of basis in the Gram determinant gives

$$v_n = \frac{K_{n+1}}{K_n}.$$

Multiplication by X is symmetric for this bilinear form. Orthogonality and degree therefore imply a three-term recurrence

$$\pi_{n+1} = (X - \beta_n)\pi_n - b_n\pi_{n-1}, \quad b_n = \frac{v_n}{v_{n-1}} = \frac{K_{n-1}K_{n+1}}{K_n^2} \quad (n \geq 1). \quad (6.2)$$

There is no analytic measure assumption.

The same recurrence gives the convergents to the formal Jacobi fraction

$$\mathcal{M} = \frac{1}{X - \beta_0 - \frac{b_1}{X - \beta_1 - \frac{b_2}{X - \beta_2 - \dots}}}. \quad (6.3)$$

For detail, let ρ_n be the polynomial part of $\mathcal{M}\pi_n$. Then $\rho_0 = 0$ and $\rho_1 = 1$. For $n \geq 1$, the coefficient of X^{-1} in $\mathcal{M}\pi_n$ is $\mathcal{L}(\pi_n) = 0$. Taking polynomial parts in (6.2) therefore shows that the ρ_n satisfy the same three-term recurrence as the π_n . Consequently ρ_n/π_n are the continued-fraction convergents in (6.3). Furthermore, all coefficients of $X^{-1}, \dots, X^{-n}$ in $\mathcal{M}\pi_n$ vanish by orthogonality. Dividing the remaining tail by the monic polynomial π_n shows that $\mathcal{M} - \rho_n/\pi_n = O(X^{-2n-1})$. Agreement to arbitrarily high order proves the formal identity. This also establishes that all polynomial partial quotients are linear. These are the classical formulas discussed in [3, Theorem 4.1]; the argument above fixes the index shift explicitly.

6.3 Polynomial complete quotients

Set

$$Z = \frac{Q}{\mathcal{M}} - P. \quad (6.4)$$

Using (5.2) and (5.5), direct substitution gives

$$Z^2 - AZ + vX + m = 0. \quad (6.5)$$

The chosen branch has $Z = A + O(X^{-2})$; its conjugate is $\bar{Z} = A - Z = O(X^{-2})$. The initial quotient is

$$F_0 = \frac{1}{\mathcal{M}} = \frac{Z + P_0}{Q_0}, \quad P_0 = P, \quad Q_0 = Q.$$

Ordinary polynomial continued-fraction division produces

$$F_n = \frac{Z + P_n}{Q_n} = a_n + \frac{1}{F_{n+1}}, \quad P_{n+1} = a_n Q_n - A - P_n, \quad (6.6)$$

with

$$Q_n Q_{n+1} = -(P_{n+1}(A + P_{n+1}) + vX + m). \quad (6.7)$$

The first norm divisibility is (5.2). Subsequent divisibility follows inductively by reducing the norm modulo Q_n and using $A + P_{n+1} \equiv -P_n \pmod{Q_n}$.

Starting with Q_0 of degree two and P_0 of degree one, the formula for P_{n+1} cancels the terms of degrees three and two, so $\deg P_{n+1} \leq 1$. The norm formula then gives $\deg Q_{n+1} \leq 2$. Since the next quotient has numerator of degree three and has a linear polynomial part, its denominator must have degree exactly two. This proves the degree assertions inductively. Write

$$P_n = d_n(X + e_n), \quad Q_n = u_n q_n, \quad q_n = X^2 + \lambda_n X + \mu_n, \quad u_0 = 1. \quad (6.8)$$

Here the u_n are local normalization constants and are unrelated to the homogeneous parameter u in theorem 2.1. Comparison of leading coefficients in (6.7) gives

$$u_{n-1}u_n = -d_n \quad (n \geq 1). \quad (6.9)$$

In particular $d_n \neq 0$. If a numerator correction had vanished in degree one, the next denominator would have degree below two and the next partial quotient would not be linear. This cannot occur in the generic moment problem just established.

The polynomial part is $a_n = (X + a - \lambda_n)/u_n$. Rescaling adjacent levels of the continued fraction with (6.9) changes its numerators to $-d_n$. Comparison with (6.3) therefore yields the essential identification

$$d_n = \frac{K_{n-1}K_{n+1}}{K_n^2} \quad (n \geq 1). \quad (6.10)$$

For example, the beginning of the rescaled fraction is

$$\mathcal{M} = \frac{1}{X + a - \lambda_0 - \frac{d_1}{X + a - \lambda_1 - \frac{d_2}{X + a - \lambda_2 - \cdots}}}.$$

There is no extra factor of 2 or 4 in this normalization, because Z , not $Y = 2Z - A$, is used throughout.

6.4 A local proof of van der Poorten's identity

Set $w = -m/v$, so $vX + m = v(X - w)$. We now derive the needed width-six identity using only polynomial division and evaluations. This is the identity in [2, Theorem 3.1], with the X^2 coefficient of A retained.

The normalized Euclidean and norm relations are

$$P_n + P_{n+1} = (X + a - \lambda_n)q_n - A, \quad (6.11)$$

$$d_n q_{n-1} q_n = P_n(A + P_n) + v(X - w). \quad (6.12)$$

Reducing the second relation modulo q_n , and using the first, shows that $P_n P_{n+1} - v(X - w)$ is divisible by q_n . Its degree is two and its leading coefficient is $d_n d_{n+1}$. Hence

$$d_n d_{n+1} q_n = P_n P_{n+1} - v(X - w). \quad (6.13)$$

This simple remainder identity is the main algebraic tool.

We first show that

$$q_n(-e_n) = -d_{n-1}. \quad (6.14)$$

For full detail, the coefficients of X^3 and X^2 in (6.12) give

$$\lambda_{n-1} + \lambda_n = a + e_n, \quad (6.15)$$

$$\lambda_{n-1}\lambda_n + \mu_{n-1} + \mu_n = b + ae_n + d_n. \quad (6.16)$$

The coefficient of X in (6.11) at index $n - 1$ is

$$d_{n-1} + d_n = \mu_{n-1} + a\lambda_{n-1} - \lambda_{n-1}^2 - b. \quad (6.17)$$

Eliminating μ_{n-1} between these equations gives

$$d_{n-1} = (a - \lambda_{n-1})e_n - \mu_n = \lambda_n e_n - e_n^2 - \mu_n = -q_n(-e_n),$$

as claimed.

Evaluate (6.13) at $X = -e_n$. Since $P_n(-e_n) = 0$, (6.14) yields

$$T_n := d_{n-1}d_nd_{n+1} = -v(e_n + w). \quad (6.18)$$

All the d_j in use are nonzero, so $e_n + w \neq 0$ in the fraction field. On the other hand, evaluating (6.13) at $X = w$ gives

$$q_n(w) = (e_n + w)(e_{n+1} + w). \quad (6.19)$$

The norm relation (6.12), also evaluated at w and divided by d_n , is

$$q_{n-1}(w)q_n(w) = (e_n + w)(A(w) + d_n(e_n + w)).$$

Substituting (6.19) and canceling the nonzero factor $e_n + w$ gives

$$(e_{n-1} + w)(e_n + w)(e_{n+1} + w) = A(w) + d_n(e_n + w). \quad (6.20)$$

Multiply by $-v^3$ and use (6.18) at three consecutive indices. The result is

$$T_{n-1}T_nT_{n+1} = v^2d_nT_n - v^3A(w), \quad (6.21)$$

or, explicitly,

$$d_{n-2}d_{n-1}^2d_n^3d_{n+1}^2d_{n+2} = v^2d_{n-1}d_n^2d_{n+1} - v^3A(w). \quad (6.22)$$

This proof explains why retaining the X^2 term in A causes no problem: only coefficient comparisons and evaluations of the full cubic have been used. For our subsequent application, $n \geq 3$, so no negative-index continued-fraction data are required.

6.5 Telescoping and removal of all denominators

Substitution of (6.10) gives two telescoping identities:

$$d_{n-1}d_n^2d_{n+1} = \frac{K_{n-2}K_{n+2}}{K_n^2}, \quad (6.23)$$

$$d_{n-2}d_{n-1}^2d_n^3d_{n+1}^2d_{n+2} = \frac{K_{n-3}K_{n+3}}{K_n^2}. \quad (6.24)$$

Multiplying (6.22) by K_n^2 and setting $N = n + 3$ gives the desired Hankel relation in $\mathcal{K}$, with coefficient $-v^3A(w)$. But

$$-v^3A(-m/v) = m^3 - avm^2 + bv^2m - cv^3 = \Gamma \quad (6.25)$$

is already polynomial. Thus both sides of (5.6) belong to $\mathcal{S}$ and are equal in its fraction field. The injection $\mathcal{S} \hookrightarrow \mathcal{K}$ proves their equality in $\mathcal{S}$ itself.

Finally, apply any ring homomorphism from $\mathcal{S}$ into a commutative ring. The specialized moments are still the unique coefficients defined by (5.4), since its triangular recursion never divided by a parameter. Determinants and all terms of (5.6) specialize as polynomials. This proves the theorem even when v, p , a normalization constant, or one or more Hankel determinants become zero. It also proves validity in positive characteristic. $\square$

7 Specialization: completion of the conjecture's proof

In the universal theorem make the substitution

$$\begin{aligned} a &= -2u, & b &= u^2 - R, & c &= -B - t, \\ p &= 0, & k &= t, & \lambda &= -u, & \mu &= 0. \end{aligned} \quad (7.1)$$

Then

$$A = X^3 - 2uX^2 + (u^2 - R)X - B - t, \quad P = t, \quad Q = X(X - u).$$

The division identity becomes

$$t(A + t) + tRX + tB = tX(X - u)^2,$$

so

$$L = t(X - u), \quad v = tR, \quad m = tB. \quad (7.2)$$

Putting these expressions into (5.4) gives precisely (2.2).

The invariant (5.7) specializes to

$$\begin{aligned} \Gamma &= t^3(B^3 + 2uRB^2 + (u^2 - R)R^2B + (B + t)R^3) \\ &= t^3(B^3 + 2uRB^2 + u^2R^2B + tR^3) \\ &= t^3(B(B + uR)^2 + tR^3). \end{aligned} \quad (7.3)$$

Equation (5.6) is therefore (2.3). This proves theorem 2.1, including the exceptional specialization $p = 0$. Using (2.4) proves theorem 1.1.

To compare with the exact expression printed by Barry, expansion of the bracket in (1.4) gives

$$\begin{aligned} \gamma/t^3 &= r^3t + r^2(s + 7t) + 2r(s^2 + 2(t + 1)s + t(t + 8)) \\ &\quad + s^3 + s^2(3t + 4) + s(t + 2)(3t + 2) + t(t^2 + 4t + 12). \end{aligned} \quad (7.4)$$

The notation on the left means the polynomial bracket, not a required division when $t = 0$. This is the coefficient in Conjecture 7. Thus the result is a proof of the proposed formula, not a nearby recurrence with adjusted parameters.

For another normalization check, the first complete quotient in the smaller family has polynomial part $X - u$ and

$$P_1 = RX + B, \quad Q_1 = -(RX + B)(X - u).$$

Its leading numerator coefficient is $d_1 = R$, exactly as required by $H_0H_2/H_1^2 = R$. This check is not needed for the specialization proof, but it catches possible sign and index errors in the curve identification.

8 Examples, singular cases, and a source correction

8.1 A positive moment sequence with signed Hankel determinants

For $r = s = t = 1$,

$$G(x) = 1 + x + 4x^2 + 12x^3 + 34x^4 + 103x^5 + 309x^6 + 934x^7 + \dots$$

The determinants, with the empty determinant included, start as

$$1, 1, 3, -10, -77, -807, 437, 468332, 2665431, 183243907, \dots$$

The coefficients are $(\alpha, \gamma) = (9, 77)$. The first two instances of (1.3) are checked by

$$\begin{aligned} 437 &= 9(-807) + 77(-10)^2, \\ 468332 &= 9 \cdot 437 \cdot 3 + 77(-77)^2. \end{aligned}$$

These checks illustrate the normalization; the theorem, not these examples, proves every subsequent relation.

Table 1: Some original-family examples; entries are exact direct determinants.

r	s	t	α	γ	H_2	H_3	H_4
0	0	1	4	17	2	-3	-17
1	1	1	9	77	3	-10	-77
-3	-2	1	1	-5	-1	-2	5
-2	-2	1	0	-1	0	-1	1
2	2	2	64	3072	4	-32	-768

8.2 What happens when a determinant vanishes

At $(r, s, t) = (-2, -2, 1)$, the directly computed beginning is

$$(H_N)_{N \geq 0} = 1, 1, 0, -1, 1, 0, -1, -1, 0, 1, -1, \dots,$$

and the identity is

$$H_N H_{N-6} = -H_{N-3}^2.$$

For $N = 8$, the expression $H_8 = (-H_5^2)/H_2$ would be $0/0$ and is not a recurrence that can be evaluated by division. The polynomial identity is nevertheless the valid equation $0 = 0$. The determinants supply a canonical sequence through this singularity; the bilinear relation alone need not uniquely determine a continuation from singular initial data.

Likewise, when $R = 0$ in the homogeneous family, $H_2 = 0$ and $w = -B/R$ cannot be used, but the final coefficients $\alpha = 0$ and $\gamma = t^3 B^3$ are perfectly well defined. The polynomial proof is doing real work here.

When $t = 0$, $F = (1 - ux)/D$ is rational. Its moments satisfy an order-at-most-three constant-coefficient recurrence from index 3 onward. Every fourth Hankel column is a linear combination of the preceding three, so $H_N = 0$ for $N \geq 4$. Both Somos coefficients also vanish, consistently with the theorem.

8.3 A parameter typo in Barry's Example 8

The source labels Example 8 with $(r, s, t) = (-2, -2, 1)$ but prints a moment sequence beginning $1, 1, 0, -3, -7, -9, \dots$ and coefficients $(\alpha, \gamma) = (1, -5)$ [1, p. 7]. Those data instead correspond to $(r, s, t) = (-3, -2, 1)$. The discrepancy is already visible in

$$g_2 = r + 3 :$$

it equals 1 at $r = -2$ and 0 at $r = -3$. Direct calculation at the corrected parameters gives

$$(H_N)_{N \geq 0} = 1, 1, -1, -2, 5, 17, -3, -122, 1201, -2980, \dots,$$

which matches the source's displayed Hankel transform after removing H_0 . This is an example-label correction, not a counterexample to Conjecture 7. The coefficient identity itself is correct.

9 Consequences and simple transformations

9.1 Integrality without a Laurent-phenomenon hypothesis

For integer parameters the moments are integers by (3.1), and so are the Hankel determinants. Thus the quotient Somos recurrence, whenever defined, produces integers along this determinant family. No separate Laurent-phenomenon theorem, or assumption that all six initial values are units, is required for this conclusion. Integrality here is immediate from the determinant representation and should not be advertised as a new integrality theorem for arbitrary Somos-6 initial data.

9.2 Binomial translations preserve the entire Hankel sequence

Let z be a new parameter and define

$$\tilde{f}_n = \sum_{j=0}^n \binom{n}{j} z^{n-j} f_j, \quad \tilde{F}(x) = \frac{1}{1-zx} F\left(\frac{x}{1-zx}\right).$$

The corresponding $N \times N$ Hankel matrices are related by

$$\tilde{\mathcal{H}}_N = T_N \mathcal{H}_N T_N^\top, \quad (T_N)_{ij} = \begin{cases} \binom{i}{j} z^{i-j}, & j \leq i, \\ 0, & j > i. \end{cases}$$

To check an entry, use the Vandermonde identity $\sum_{p+q=j} \binom{i}{p} \binom{k}{q} = \binom{i+k}{j}$. Since T_N has diagonal entries 1, all Hankel determinants are unchanged. Consequently the same Somos recurrence holds for every binomial translate. This is a standard Hankel invariance; the short proof is included to show exactly what happens to this family.

9.3 Scaling and weights

Replacing f_n by $\rho^n f_n$ multiplies H_N by $\rho^{N(N-1)}$, by factoring ρ^i from row i and ρ^j from column j . The two recurrence coefficients therefore scale by ρ^{10} and ρ^{18} , respectively. In theorem 2.1, the parameter weights are

$$\text{wt}(u) = 1, \quad \text{wt}(R) = 2, \quad \text{wt}(B) = \text{wt}(t) = 3.$$

Both coefficient formulas have exactly the predicted weights. This provides an additional structural check on the exponents in the result.

More explicitly, for invertible u , the homogeneous series is $F(x) = G(ux)$ with the original parameters

$$r = R/u^2 - 2, \quad s = (B - t)/u^3, \quad \tau = t/u^3$$

in place of (r, s, t) . The polynomial theorem additionally covers $u = 0$, where this rescaling is unavailable.

10 Reproducible exact computations

The mathematical proof is symbolic and valid for all indices. Computation was used to select the target, detect normalization errors, and audit the finished formulas. It is deliberately independent of the proposed Somos recurrence.

10.1 What the programs actually do

The standard-library program code/`verify.py` generates moments from the quadratic defining equation, constructs each Hankel matrix separately, and evaluates its determinant by fraction-free Bareiss elimination with row pivoting. Every integer division is checked for exactness. It never generates a test determinant from the Somos recurrence. For rational parameter tests, exact Fraction arithmetic is used. Small determinant computations are also checked against the permutation definition of the determinant.

The full deterministic run uses seed 20260919 and the following test groups:

Group	Parameter cases	Maximum matrix size
Original family, $(r, s, t) \in \{-3, \dots, 3\}^3$	343	14
Homogeneous family, $(u, R, B, t) \in \{-2, \dots, 2\}^4$	625	14
Seven-parameter polynomial family, seeded integers	128	14
Larger-matrix tests in the polynomial family	16	24
Homogeneous family with rational parameters	16	10
Total	1128	24

The saved run reports 10,248 zero recurrence residuals in these cases. Among the 1,128 cases, 553 contain at least one zero Hankel determinant of size at least two; there are 3,465 such zero determinant values in the test records. Thus the test set does not silently exclude the singular specializations that require care in the proof. There are also 84 small-matrix determinant cross-checks, 343 independent original-family moment-recursion comparisons, and 25 Catalan-specialization determinants $H_0, \dots, H_{24}$, all equal to 1.

Five additional named examples are recorded through H_{20} and through moment g_{38} , adding 75 recurrence checks. These are reported separately from the 10,248 main-suite residuals. The exact output and all parameters are in `data/verification_summary.json` and `data/verification_cases.json`; the example values are also supplied as CSV files.

10.2 Finite symbolic certificates

The optional program code/`symbolic_certificates.py` uses SymPy to verify 16 explicit algebraic identities. They include the square completion, the initial norm factorization, the equality of the expanded and compact gamma formulas, the universal polynomial division, all essential specialization formulas, the coefficient elimination proving $q_n(-e_n) = -d_{n-1}$, both Hankel telescoping identities, and the symbolic initial determinants H_2, H_3, H_4 . The saved run used SymPy 1.14.0.

These finite checks are not described as a proof of an infinite recurrence. The all-index step is the local polynomial identity in Section 6.4, followed by telescoping and polynomial specialization. No Lean or other proof-assistant verification is claimed.

10.3 Commands and complexity

From the archive root, run

```
python code/verify.py
python code/symbolic_certificates.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The first command requires only Python 3.10 or later. The second requires SymPy; an optional requirements file is included. A smaller numerical run is available with `python code/verify.py -quick`.

The direct verification method uses $O(N^2)$ arithmetic operations to obtain moments through order $2N$ by a quadratic convolution recursion, and $O(N^3)$ operations for one $N \times N$ determinant. Computing every $H_0, \dots, H_N$ separately therefore takes $O(N^4)$ arithmetic operations. These are operation counts, not bit-complexity estimates; exact integers can grow rapidly. Once the theorem is proved, a nonzero Hankel sequence can be extended much faster by its Somos recurrence, but doing that in the verifier would make the numerical evidence circular. Singular cases should continue to be evaluated from moments and determinants unless a separate justified singularity-handling method is supplied.

11 Conclusion and scope of the result

The selected conjecture is resolved by the polynomial identity (1.3). The key reduction is not an accidental match of initial terms: the sextic discriminant is exactly a cubic square plus a linear remainder, and its two curve invariants reproduce the conjectured coefficients for every parameter value. The seven-parameter deformation provides a rigorous route from generic continued fractions to exceptional initial data and zero determinants.

The result does not prove Barry's Somos-8 conjectures, classify arbitrary Somos-6 initial-value problems, establish positivity of the Hankel polynomials, or assert that all sequences satisfying the bilinear relation have this determinant representation. Those would be separate questions. The principal mathematical ingredient belongs to the existing work of van der Poorten and the established Hankel/continued-fraction correspondence. What this manuscript adds is an explicit application with every normalization and specialization step made checkable.

Independent expert review remains appropriate before making a publication or priority claim. The archive is designed to facilitate that review: it contains the editable proof, its compiled version, a bounded literature status note, and reproducible computations rather than only an assertion that a conjecture has been solved.

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